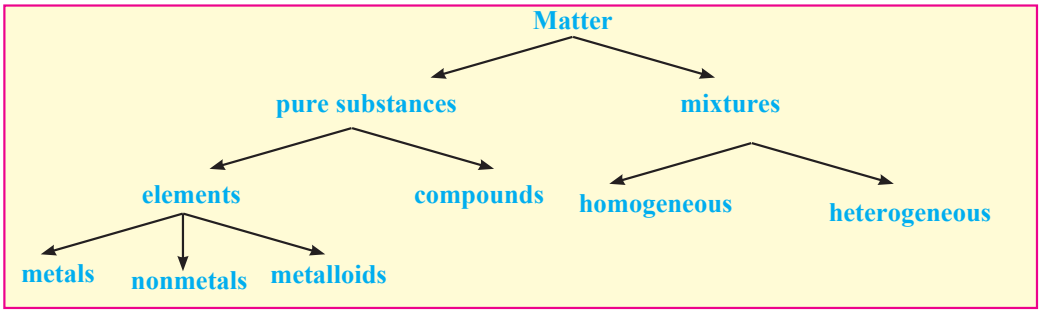
1. Some Basic Concepts of Chemistry

1.1 Introduction : Chemistry is the study of matter, its physical and chemical properties and the physical and chemical changes it undergoes under different conditions.

1.2.1. Matter : You have learnt earlier that matter occupies space and has mass. **Matter** can be further **classified** into pure substances and mixtures **on the basis of chemical composition**.



Chemistry is a central science. Its knowledge is required in the studies of physics, biological sciences, applied sciences, and earth and space sciences. The scope of chemistry is in every aspect of life, for example, the air we breathe, the food we eat, the fluids we drink, our clothing, transportation and fuel supplies, and so on.

Though it is an ancient science, due to development and advancement in science and technology, chemistry has developed as modern science. Technological development in sophisticated instruments expanded our knowledge of chemistry which, now, has been used in applied sciences such as medicine, dentistry, engineering, agriculture and in daily home use products.

1.2 Nature of Chemistry : Chemistry is traditionally classified further into five branches : organic, inorganic, physical, bio and analytical. Organic chemistry is the study of the properties and reactions of compounds of carbon. Inorganic chemistry is the study of all substances which are not organic. Physical chemistry is the study of principles underlying chemistry. It deals with the studies of properties of matter. It is study of atoms, molecules, and fundamental concepts related to electrons, energies and dynamics therein. It provides basic framework for all the other branches of chemistry.

Let us understand first what are pure substances and mixtures.

1.2.2 Pure substances versus mixtures :

Pure substances have a definite chemical composition. They always have the same properties regardless of their origin. Mixtures have no definite chemical composition and hence no definite properties.

Examples of pure substances : Pure metal, distilled water, etc.

Examples of mixtures : Paint (mixture of oils, pigment, additive), concrete (a mixture of sand, cement, water)

Can you tell?

Which are mixtures and pure substances from the following ?
i. sea water ii. gasoline iii. skin iv. a rusty nail v. a page of the textbook. vi. diamond

Pure substances are further divided into elements and compounds. **Elements are pure substances which can not be broken down into simpler substances by ordinary chemical changes.** Elements are further classified as metals, nonmetals and metalloids.

1. Metals :
i. have a lustre (a shiny appearance).
ii. conduct heat and electricity.
iii. can be drawn into wire (are ductile).
iv. can be hammered into thin sheets (are malleable).
Examples : gold, silver, copper, iron. Mercury is a liquid metal at room temperature.

2. Non-metals :
i. have no lustre. (exception : diamond, iodine)
ii. are poor conductors of heat and electricity. (exception : graphite)
iii. can not be hammered into sheets or drawn into wire, because they are brittle.
Examples : Iodine, nitrogen, carbon, etc.

3.Metalloids : Some elements have properties intermediate between metals and non-metals and are called metalloids or semi-metals. Examples include arsenic, silicon and germanium.

Compounds are the pure substances which can be broken down into simpler substances by ordinary chemical changes. In a compound, two or three elements are combined in a fixed proportion.

Mixture contains two or more substances in no fixed proportions and may be separated by physical methods. Mixtures are further divided into homogeneous and heterogeneous. Solutions are homogeneous mixtures, because the molecules of constituent solute and solvent are uniformly mixed throughout its bulk. In heterogeneous mixtures the molecules of the constituents are not uniformly mixed throughout the bulk. For example : Suspension of an insoluble solid in a liquid.

1.2.3 States of matter : You are also aware that matter exists in three different states namely gas, liquid and solid. You are going to learn about these states in unit 3 (chapter 10).

In solids, constituent atoms or molecules (particles) are tightly held in perfect order and therefore solids possess definite shape and volume. Liquids contain particles close to each other and they can move around within the liquid. While in gases, the particles are far apart as compared to those in liquid and solid state.

Three states of matter are interconvertible by changing the conditions of temperature and pressure.

Can you tell?

Classify the following as element and compound.
i. mercuric oxide ii. helium gas iii. water iv. table salt v. iodine vi. mercury vii. oxygen viii. nitrogen

1.3 Properties of matter and their measurement :



Fig : 1.1 Burning of magnesium wire

Different kinds of matter have characteristic properties, which can be classified into two categories as **physical properties** and **chemical properties**.

Physical properties are those which can be measured or observed without changing the chemical composition of the substance. Colour, odour, melting point, boiling point, density, etc. are physical properties. Chemical properties are the properties where substances undergo a chemical change and thereby exhibit change in chemical composition. For example, coal burns in air to produce carbon dioxide or magnesium wire burns in air in the presence of oxygen to form magnesium oxide. (Fig. 1.1)

1.3.1 Measurement of properties : Many properties of matter are quantitative in nature. When you measure something, you are comparing it with some standard. The standard quantity is reproducible and unchanging.

Many properties of matter such as mass, length, area, pressure, volume, time, etc. are quantitative in nature. Any quantitative measurement is expressed by a number followed by units in which it is measured. For example, length of class room can be represented as 10 m. Here 10 is the number and 'm' denotes metre-the unit in which the length is measured.

The standards are chosen orbitrarily with some universally acceped criteria. **"The arbitrarily decided and universally accepted standards are called units."**

There are several systems in which units are expressed such as CGS (centimetre for length, gram for mass and second for time), FPS (foot, pound, second) and MKS (metre, kilogram, second) systems, etc.

SI units :

In 1960, the general conference of weights and measure, proposed revised metric system, called **International System of units,** that is, **SI units**.

The metric system which originated in France in late eighteenth century, was more convenient as it was based on the decimal system. Later, based on a common standard system, the International System of Units (SI units) was established.

The SI system has seven base units as listed in Table 1.1. These are fundamental scientific quantities. Other units like speed, volume, density, etc. can be derived from these quantities.

Table 1.1 SI Fundamental units

Base Physical Quantity	Symbol for Quantity	Name of SI Unit	Symbol for SI Unit
Length	l	metre	m
Mass	m	kilogram	kg
Time	t	second	s
Electric current	I	ampere	A
Thermodynamic temperature	T	Kelvin	K
Amount of substance	n	mole	mol
Luminous intensity	I_v	candela	cd

1.3.2 Physical properties

i. Mass and weight : We know that matter has mass. So mass is an inherent property of matter. It is the measure of the quantity of matter a body contains. The mass of a body does not vary as its position changes. On the other hand, the weight of a body is result of the mass and gravitational attraction. The weight of a body varies because the gravitational attraction of the earth for a body varies with the distance from the centre of the earth.

Hence, **the mass of a body is more fundamental property than its weight**.

The basic unit of mass in the SI system is the kilogram as given in Table 1.1. However, a fractional quantity 'gram' is used for weighing small quantities of chemicals in the laboratories. Therefore, in terms of grams it is defined (1kg = 1000 g = 10^3 g)

ii. Length : In chemistry we come across 'length' while expressing properties such as the atomic radius, bond length, wavelenght of electromagnetic radiation, and so on. These quantities are very small therefore fractional units of the SI unit of length are used for example, nanometre (nm), picometre (pm). Here 1nm = 10^{-9} m, 1 pm = 10^{-12} m.

iii. Volume : It is the amount of space occupied by a three dimensional object. It does not depend on shape. For measurement of volume of liquids and gases, a common unit, litre (L) which is not an SI unit is used.

1 L = 1 dm^3 = 1000 mL = 1000 cm^3

1000 cm^3 = 10 cm × 10 cm × 10 cm of volume

SI unit of volume is expressed as $(metre)^3$ or m^3.

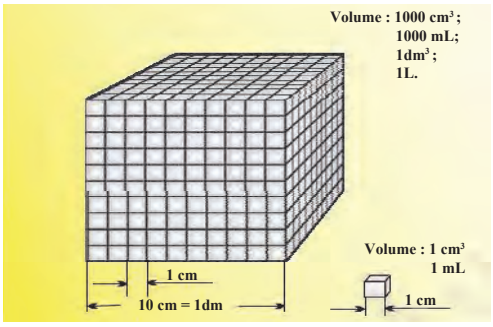


Fig. 1.2 : Litre and SI unit of volume

Different kinds of glassware are used to measure the volume of liquids and solutions. For example, graduated cylinder, burette, pipette, etc. A volumetric flask is used to prepare a known volume of a solution. Figure 1.3 shows the types of apparatus used in laboratory for measuring volume of liquids.

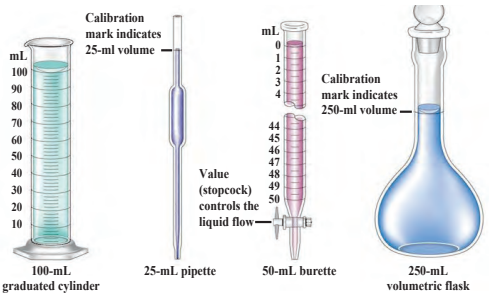


Fig. 1.3 : Volumetric glass apparatus

iv. Density : Density of a substance is its mass per unit volume. It is determined in the laboratory by measuring both the mass and the volume of a sample. The density is calculated by dividing mass by volume. It is the characteristic property of a substance.

So SI unit of density can be obtained as follows:

$$\text{SI unit of density} = \frac{\text{SI unit of mass}}{\text{SI unit of volume}}$$

$$= \frac{\text{kg}}{\text{m}^3} \text{ or kg m}^{-3}$$

CGS units it is $\frac{\text{g}}{\text{mL}}$ or g mL^{-1} or g cm^{-3}

v. Temperature : Temperature is a measure of the hotness or coldness of an object. There are three common scales to measure temperature, namely ^{0}C (degree Celsius), ^{0}F (degree Fahrenheit) and K (Kelvin). Here K is the SI unit. Figure 1.4 shows the thermometers based on these scales.

Generally, the thermometer with celsius scale are calibrated from 0 ^{0}C to 100 ^{0}C where these two temperatures are respectively the freezing point and the boiling point of water at atmospheric pressure. These are represented on fahrenheit scale as 32^0 F to 212^0 F.

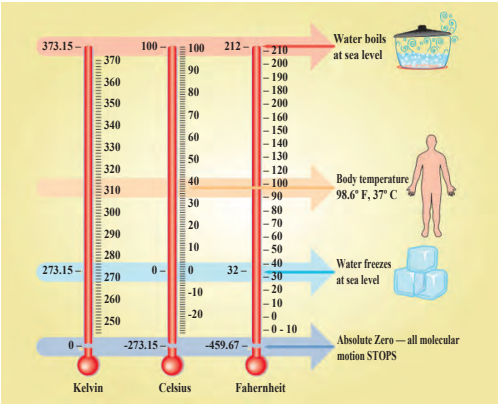


Fig 1.4 : Thermometers of different temperature scale

The temperatures on two scales are related to each other by the following relationship :

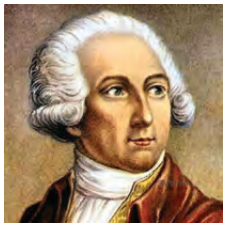
$$^0\text{F} = \frac{9}{5}(^0\text{C}) + 32$$

The Kelvin scale is related to Celsius scale as follows :

$$\text{K} = ^0\text{C} + 273.15$$

1.4 Laws of Chemical Combination : The elements combine with each other and form compounds. This process is governed by five basic laws discovered before the knowledge of molecular formulae.

1.4.1 Law of conservation of mass : Antoine



Lavoisier (1743-1794) a French scientist is often referred to as the father of modern chemistry. He carefully performed many combustion experiments, namely, burning of phosphorus and mercury, both in the presence of air. Both resulted in an increase in weight. After several experiments he found that the weight gained by the phosphorus was

exactly the same as the weight lost by the air. He observed that,

Total mass of reactants

= Total mass of products

When hydrogen gas burns and combines with oxygen to yield water, the mass of the water formed is equal to the mass of the hydrogen and oxygen consumed. Thus, the law of conservation of mass states that **'mass can neither be created nor destroyed.'**

1.4.2 Law of Definite Proportions :

French chemist, Joseph Proust performed experiments on two samples of cupric carbonate. One of the samples was natural in origin and the other was a synthetic one. He found that the composition of elements present in it was same for both the samples as shown below :

Cupric Carbonate	% of copper	% of oxygen	% of carbon
Natural sample	51.35	38.91	9.74
Synthetic sample	51.35	38.91	9.74

This led Joseph Proust to state the law of definite proportion as follows :

'A given compound always contains exactly the same proportion of elements by weight.' Irrespective of the source, a given compound always contains same elements in the same proportion. The validity of this law has been confirmed by various experiments. This law is sometimes referred to as **Law of definite composition**.

1.4.3 Law of multiple proportions :

This law was proposed by John Dalton in 1803. It has been observed that two or more elements may form more than one compound. Law of multiple proportions summarizes many experiments on such compounds. **When two elements A and B form more than one compounds, the masses of element B that combine with a given mass of A are always in the ratio of small whole numbers.** For example, **i.** Hydrogen combines with oxygen to form two compounds, namely water and hydrogen peroxide.

Hydrogen + Oxygen ⟶ Water

2 g 16 g 18 g

Hydrogen + Oxygen ⟶ Hydrogen Peroxide

2 g 32 g 34 g

Here, it is found that, the two masses of oxygen i.e. 16 g and 32 g which combine with a fixed mass of hydrogen (2g) are in the ratio of small whole numbers, i. e. 16:32 or 1:2.

ii. Nitrogen and oxygen combine to form two compounds, nitric oxide and nitrogen dioxide.

Nitrogen + Oxygen ⟶ Nitric Oxide

14 g 16 g 30 g

Nitrogen + Oxygen ⟶ Nitrogen Dioxide

14 g 32 g 46 g

Here, you find that the two masses of oxygen i.e. 16 g and 32 g when combine with a fixed mass of Nitrogen (14 g) are in the ratio of small whole numbers i.e. 16:32 or 1:2.

(Similar examples such as CO and CO_2 (1:2 ratio), SO_2 and SO_3 (2:3 ratio), can be found.)

1.4.4 Gay Lussac Law of Gaseous Volume :

This law was put forth by Gay Lussac in 1808. The law states that **when gases combine or are produced in a chemical reaction they do so in a simple ratio by volume, provided all gases are at same temperature and pressure**.



Illustration : i. Under the same conditions of temperature and pressure, 100 mL of hydrogen combines with 50 mL of oxygen to give 100 mL of water vapour.

Hydrogen (g) + Oxygen (g) ⟶ Water(g)

100 mL 50 mL 100 mL

(2 vol) (1 vol) (2 vol)

Thus, the volumes of hydrogen gas and oxygen gas which combine together i.e. 100 mL and 50 mL producing two volumes of water vapour which amounts to 100 mL bear a simple ratio of 2:1:2

ii. Under the same condition of temperature and pressure,

1 L of nitrogen gas combines with 3 L of hydrogen gas to produce 2 L of ammonia gas.

Nitrogen (g) + Hydrogen (g) ⟶ Ammonia(g)

1 L 3 L 2 L

(1 vol) (3 vol) (2 vol)

Thus, the volume of nitrogen gas and hydrogen gas which combine together i.e. 1 L and 3 L and volume of ammonia gas produced i. e. 2 L bear a simple ratio of 1:3:2.

Remember

Gay Lussac's discovery of integer ratio in volume relationship is actually the law of definite proportion by gaseous volumes.

Can you tell?

If 10 volumes of dihydrogen gas react with 5 volumes of dioxygen gas, how many volumes of water vapour would be produced?



1.5 Avogadro Law : In 1811, Avogadro proposed that **equal volumes of all gases at the same temperature and pressure contain equal number of molecules**.

If we consider the reaction of hydrogen and oxygen to produce water vapour.

Hydrogen (g) + Oxygen (g) ⟶ Water (g)

100 mL 50 mL 100 mL

(2 vol) (1 vol) (2 vol)

(Gay Lussac Law)

2n molecules n molecules 2n molecules

(Avogadro law)

2 molecules 1 molecule 2 molecules

We see that 2 volumes of hydrogen combine with 1 volume of oxygen to give 2 volumes of water vapour, without leaving any unreacted oxygen. According to Avogadro law, if 1 volume contains n molecules, then 2n molecules of hydrogen combine with n molecules of oxygen to give 2n molecules of water.

Therefore, 2 molecules of hydrogen gas combine with 1 molecule of oxygen to give 2 molecules of water vapour. Avogadro could explain the above result by considering the molecules to be polyatomic. If hydrogen and oxygen were considered as diatomic, as recognized now, then the above results are easily understandable.

Remember

Avogadro made a distinction between atoms and molecules, which is quite understandable in the present time.

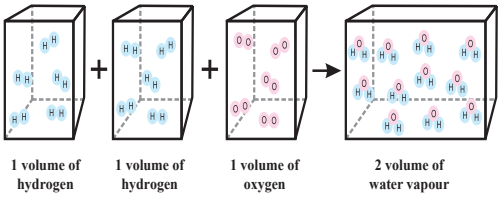


Fig. 1.5 : two volume of hydrogen react with one volume of oxygen to give two volumes of water vapour

1.6 Dalton's Atomic Theory : In 1808, Dalton published "A New System of chemical philosophy" in which he proposed the following features, which later became famous as Dalton's atomic theory.

1. Matter consists of **tiny, indivisible particles** called **atoms**.
2. All the atoms of a given elements have **identical** properties including **mass**. Atoms of different elements differ in mass.
3. **Compounds** are formed when atoms of different elements combine in a fixed ratio.
4. **Chemical reactions** involve only the **reorganization of atoms**. Atoms are neither created nor destroyed in a chemical reaction.

Dalton's theory could explain all the laws of chemical combination.

Can you recall?

What is an atom and a moleule ? What is the order of magnitude of mass of one atom ? What are isotopes?

1.7 Atomic and molecular masses : You know about the terms atoms and molecules. Thus it is appropriate here to understand what we mean by atomic and molecular masses.

1.7.1 Atomic Mass : Every element has a characteristic atomic mass. Atomic mass is the mass of an atom. It is actually very very small. For example, the mass of one hydrogen atom is 1.6736×10^{-24} g. This is very small quantity and not easy to measure.

In the present system, mass of an atom is determined relative to the mass of a carbon - 12 atom as the standard and this has been agreed upon in 1961 by IU PAC. In this system, an atom of carbon-12 is assigned a mass of exactly 12.00000 atomic mass unit **(amu)** and all other atoms of other elements are given a relative atomic mass, to that of carbon - 12. The atomic masses are expressed in **amu**. **One amu is defined** as a **mass exactly equal to one twelth of the mass of one carbon-12 atom**. Later on the exact value of atomic mass unit in grams was experimentally established.

$$1 \text{ amu} = \frac{1}{12} \times \text{mass of one C-12}$$

$$= \frac{1}{12} \times 1.992648 \times 10^{-23} \text{ g}$$

$$= 1.66056 \times 10^{-24} \text{ g}$$

Recently, amu has been replaced by **unified mass unit** called **dalton** (symbol '**u**' or '**Da**'), 'u' means unified mass.

Problem 1.1 : Mass of an atom of oxygen in gram is 26.56896×10^{-24} g. What is the atomic mass of oxygen in u ?

Solution : Mass of an atom of oxygen in gram is 26.56896×10^{-24} g, and

1.66056×10^{-24} g = 1 u

$\therefore$ 26.56896×10^{-24} g = ?

$$= \frac{26.56896 \times 10^{-24} \text{ g}}{1.66056 \times 10^{-24} \text{ g/u}} = 16.0 \text{ u}$$

Similarly mass of an atom of hydrogen = 1.0080 u

1.7.2 Average Atomic Mass : Many naturally occuring elements exist as mixture of more than one isotope. Isotopes have different atomic masses. The atomic mass of such an element is the weighted average of atomic masses of its isotopes (taking into account the atomic masses of isotopes and their relative abundance i.e. percent occurrance). This is called **average atomic mass** of an element. For example, carbon has the following three isotopes with relative abundances and atomic masses as shown against each of them.

Isotope	Atomic mass (u)	Relative Abundance (%)
^{12}C	12.00000	98.892
^{13}C	13.00335	1.108
^{14}C	14.00317	2×10^{-10}

From the above data, the average atomic mass of carbon

= (12 u) (98.892/100) + (13.00335 u) (1.108/100) + (14.00317) (2×10^{-10}/100)

= 12.011 u

Similary, average atomic masses for other elements can be calculated.



Remember

- In the periodic table of elements, the atomic masses mentioned for different elements are actually their average atomic masses.
- For practical purpose, the average atomic mass is rounded off to the nearest whole number when it differs from it by a very small fraction.

Element	Isotopes	Average atomic mass	Rounded off atomic mass
Carbon	^{12}C, ^{13}C, ^{14}C	12.011 u	12.0 u
Nitrogen	^{14}N, ^{15}N	14.007 u	14.0 u
Oxygen	^{16}O, ^{17}O, ^{18}O	15.999 u	16.0 u
Chlorine	^{35}Cl, ^{37}Cl	35.453 u	35.5 u
Bromine	^{79}Br, ^{81}Br	79.904 u	79.9 u

Problem 1.2 : Calculate the average atomic mass of neon using the following data :

Isotope	Atomic mass	Natural Abundance
^{20}Ne	19.9924 u	90.92%
^{21}Ne	20.9940 u	0.26 %
^{22}Ne	21.9914 u	8.82 %

Solution : Average atomic mass of Neon (Ne)

$$= \frac{\text{Atomic mass of } ^{20}Ne \times \% + \text{Atomic mass of } ^{21}Ne \times \% + \text{Atomic mass of } ^{22}Ne \times \%}{100}$$

$$= \frac{(19.9924u)(90.92) + (20.9940u)(0.26) + (21.9914u)(8.82)}{100} = 20.1707 \text{ u}$$

1.7.3 Molecular Mass : Molecular mass of a substance is the sum of average atomic masses of all the atoms of elements which constitute the molecule. Molecular mass of a substance is the mass of one molecule of that substance relative to the mass of one carbon-12 atom. It is obtained by multiplying average atomic mass of each element by the number of its atoms and adding them together.

For example, the molecular mass of carbon dioxide (CO_2) is

= 1(average atomic mass of C)
+ 2 (average atomic mass of O)
= 1 (12.0 u) + 2 (16.0 u) = 44.0 u

Some more examples of calculations of molecular mass.

i. $H_2O = 2 \times 1 \text{ u} + 16 \text{ u} = 18 \text{ u}$

ii. $C_6H_5Cl = (6 \times 12 \text{ u}) + (5 \times 1 \text{ u}) + (35.5 \text{ u})$
$= 112.5 \text{ u}$

iii. $H_2SO_4 = (2 \times 1 \text{ u}) + (32 \text{ u}) + (4 \times 16 \text{ u}) = 98 \text{ u}$

Problem 1.3 : Find the mass of 1 molecule of oxygen (O_2) in amu (u) and in grams.

Solution : Molecular mass of $O_2 = 2 \times 16$ u

$\therefore$ mass of 1 molecule = 32 u

$\therefore$ mass of 1 molecule of O_2
$= 32.0 \times 1.66056 \times 10^{-24}$ g
$= 53.1379 \times 10^{-24}$ g

1.7.4 Formula Mass

Some substances such as sodium chloride do not contain discrete molecules as the constituent units. In such compounds, cationic (sodium) and anionic (chloride) entities are arranged in a three dimensional structure.

In sodium chloride crystal, one $Na^{\oplus}$ ion is surrounded by six $Cl^{\ominus}$ ions, all at the same distance from it and vice versa. Therefore, NaCl is the formula used to represent sodium chloride, though it is not a molecule. Similarly, a term 'formula mass' is used for such ionic compounds, instead of molecular mass. **The formula mass of a substance is the sum of atomic masses of the atoms present in the formula.**

Problem 1.4 : Find the formula mass of
i. NaCl ii. $Cu(NO_3)_2$

i. Formula mass of NaCl
= average atomic mass of Na
+ average atomic mass of Cl
= 23.0 u + 35.5u = 58.5 u

ii. Formula mass of $Cu(NO_3)_2$
= average atomic mass of Cu + 2 × (average atomic mass of nitrogen + average atomic mass of three oxygen)
= (63.5) + 2(14 + 3 x 16) = 187.5 u



Try this

Find the formula mass of $CaSO_4$
If atomic mass of Ca = 40.1 u,
S =32.1 u and O = 16.0 u

1.8 Mole concept and molar mass



Can you recall?

1. One dozen means how many items ?
2. One gross means how many items ?

Mole : Expressing large count of objects is made easy by using quantitative adjectives such as dozen, gross. You know that even a small amount of any substance contains very large number of atoms or molecules. We use a quantitative adjective 'mole' to express the large number of submicroscopic entities like atoms, ions, electrons, etc. present in a substance.

Definiton : One mole is the amount of a substance that contains as many entities or particles as there are atoms in exactly 12 g (or 0.012 kg) of the carbon -12 isotope.

Let us calculate the number of atoms in 12.0000 g of Carbon-12 isotopes. Mass of one carbon-12 atom (determined by mass spectrometer) = 1.992648×10^{-23} g,

Mass of one mole carbon atom = 12 g

$\therefore$ Number of atoms in 12 g of carbon -12

$$= \frac{12 \text{g/mol}}{1.992648 \times 10^{-23} \text{ g/atom}}$$

$$= 6.0221367 \times 10^{23} \text{ atom/mol}$$

Thus one mole is the amount of a substance that contains 6.0221367×10^{23} particles/ entities (such as atoms, molecules or ions).

Note that the name of the unit is **mole** and the symbol for the unit is **mol**.

Remember

The number 6.0221367×10^{23} is known as **Avogadro's Constant 'N_A'** in the honour of Amedo Avogadro.

In SI system, mole (Symbol mol) was introduced as seventh base quantity for the amount of a substance.

Example :

1 mole of oxygen atoms = 6.0221367×10^{23} atoms of oxygen

1 mole of water molecules = 6.0221367×10^{23} molecules of water

1 mole of sodium chloride = 6.0221367×10^{23} formula units of NaCl

Molar Mass : The mass of one mole of a substance (element/compound) in grams is called its molar mass. The molar mass of any element in grams is numerically equal to atomic mass of that element in u.

Element	Atomic mass (u)	Molar mass (g mol^{-1})
H	1.0 u	1.0 g mol^{-1}
C	12.0 u	12.0 g mol^{-1}
O	16.0 u	16.0 g mol^{-1}

Simillary molar mass of any substance, existing as polyatomic molecule, in grams is numerically equal to its molecular mass or formula mass in u.

Polyatomic substance	Molecular/ formula mass (u)	Molar mass (g mol^{-1})
O_2	32.0 u	32.0 g mol^{-1}
H_2O	18.0 u	18.0 g mol^{-1}
NaCl	58.5 u	58.5 g mol^{-1}

Molar mass of O atoms

$= 6.022 \times 10^{23}$atom/mol $\times$ 16 u/atom $\times 1.66056 \times 10^{-24}$ g/u $= 16.0$ g/mol

Problem 1.5 : Calculate the number of moles and molecules of urea present in 5.6 g of urea.

Solution : Mass of urea = 5.6 g

Molecular mass of urea, NH_2CONH_2

= 2 (average atomic mass of N) + 4 (average atomic mass of H) + 1(average atomic mass of C) + 1(average atomic mass of O)

$= 2 \times 14$ u $+1 \times 12$ u $+4 \times 1$ u $+ 1 \times 16$ u

$= 60$ u

$\therefore$ molar mass of urea = 60 g mol^{-1}

Number of moles

$$= \frac{\text{mass of urea in g}}{\text{molar mass of urea in g mol}^{-1}}$$

$$= \frac{5.6 \text{ g}}{60 \text{ g mol}^{-1}} = 0.0933 \text{ mol}$$

Number of molecules = Number of moles $\times$ Avogadro's constant

Number of molecules of urea

$= 0.0933 \times 6.022 \times 10^{23}$ molecules/mol

$= 0.5618 \times 10^{23}$ molecules

$= 5.618 \times 10^{22}$ molecules

Ans : Number of moles = 0.0933 mol

Number of molecules of urea

$= 5.618 \times 10^{22}$ molecules

Problem 1.6 : Calculate the number of atoms in each of the following
i. 52 moles of Argon (Ar)
ii. 52 u of Helium (He)
iii. 52 g of Helium (He)

Solution :
i. 52 moles of Argon
1 mole Argon atoms = 6.022×10^{23} atoms of Ar
$\therefore$ 52 moles of Ar

$$= 52 \text{ moles} \times \frac{6.022 \times 10^{23}}{1 \text{mol}} \text{atoms}$$

$= 313.144 \times 10^{23}$ atoms of Argon

ii. 52 u of Helium
Atomic mass of He = mass of 1 atom of He = 4.0u
$\therefore$ 4.0 u = 1 He $\therefore$ 52 u = ?

$$= 52 \text{ u} \times \frac{1 \text{atom}}{4.0 \text{ u}} = 13$$

atoms of He
iii. 52 g of He
Mass of 1 mole of He = 4.0 g

Number of moles of He

$$= \frac{\text{mass of He}}{\text{mass of 1mole of He}}$$

$$= \frac{52 \text{ g}}{4.0 \text{ g mol}^{-1}} = 13 \text{ mol}$$

Number of atoms of He
= Number of moles $\times 6.022 \times 10^{23}$
= 13 mol $\times 6.022 \times 10^{23}$ atoms/mol
= 78.286×10^{23} atoms of He.

1.9 Moles and gases : Many substances exist as gases. If we want to find the number of moles of gas, we can do this more conveniently by measuring the volume rather than mass of the gas. Chemists have deduced from Avogardro law that "One mole of any gas occupies a volume of 22.4 dm^3 at **standard temperature (0^0C) and pressure (1 atm) (STP)**. The volume of 22.4 dm^3 at STP is known as molar volume of a gas.

Number of moles of a gas (n) =

$$\frac{\text{Volume of the gas at STP}}{\text{Molar volume of gas}}$$

Thus
Number of moles of a gas (n) =

$$\frac{\text{Volume of the gas at STP}}{22.4 \text{ dm}^3\text{mol}^{-1}}$$

Number of molecules = number of moles $\times$ 6.022×10^{23} molecules mol^{-1}
(Note : IUPAC has recently changed the standard pressure to 1 bar. Under these **new STP conditions** the molar volume of a gas is 22.71 L mol^{-1})

Problem 1.7 : Calculate the number of moles and molecules of ammonia (NH_3) gas in a volume 67.2 dm^3 of it measured at STP.

Solution :
Volume of NH_3 at STP = 67.2 dm^3
molar volume of a gas = 22.4 dm^3 mol^{-1}

Number of moles (n)

$$= \frac{\text{Volume of the gas at STP}}{\text{Molar volume of gas}}$$

$$\text{Number of moles of } NH_3 = \frac{67.2 \text{ dm}^3}{22.4 \text{ dm}^3 \text{ mol}^{-1}}$$

= 3.0 mol
Number of molecules = Number of moles $\times$ 6.022×10^{23} molecules mol^{-1}
Number of molecules of NH_3 = 3.0 mol $\times$ 6.022×10^{23} molecules mol^{-1}
= 18.066×10^{23} molecules



Try this

Calculate the volume in dm^3 occupied by 60.0 g of ethane at STP.

Exercises

1. Choose the most correct option.

A. A sample of pure water, whatever the source always contains ..... by mass of oxygen and 11.1 % by mass of hydrogen.
a. 88.9 b. 18 c. 80 d. 16

B. Which of the following compounds can NOT demonstrate the law of multiple proportions ?
a. NO, NO_2 b. CO, CO_2
c. H_2O, H_2O_2 d. Na_2S, NaF

C. Which of the following temperature will read the same value on celsius and Fahrenheit scales.
a. -40^0 b. $+40^0$
c. -80^0 d. -20^0

D. SI unit of the quantity electric current is .....
a. Volt b. Ampere
c. Candela d. Newton

E. In the reaction $N_2 + 3H_2 \longrightarrow 2NH_3$, the ratio by volume of N_2, H_2 and NH_3 is 1 : 3 : 2 This illustrates the law of
a. definite proportion
b. reciprocal proportion
c. multiple proportion
d. gaseous volumes

F. Which of the following has maximum number of molecules ?
a. 7 g N_2 b. 2 g H_2
c. 8 g O_2 d. 20 g NO_2

G. How many g of H_2O are present in 0.25 mol of it ?
a. 4.5 b. 18
c. 0.25 d. 5.4

H. The number of molecules in 22.4 cm^3 of nitrogen gas at STP is
a. 6.022×10^{20}
b. 6.022×10^{23}
c. 22.4×10^{20}
d. 22.4×10^{23}

I. Which of the following has the largest number of atoms ?
a. 1g Au (s) b. 1g Na (s)
c. 1g Li (s) d. 1g Cl_2 (g)

2. Answer the following questions.

A. State and explain Avogadro's law.

B. Point out the difference between 12 g of carbon and 12 u of carbon

C How many grams does an atom of hydrogen weigh ?

D. Calculate the molecular mass of the following in u.
a. NH_3 b. CH_3COOH c. C_2H_5OH

E. How many particles are present in 1 mole of a substance ?

F. What is the SI unit of amount of a substance ?

G. What is meant by molar volume of a gas ?

H. State and explain the law of conservation of mass.

I. State the law of multiple proportions.

3. Give one example of each

A. homogeneous mixture
B. heterogeneous mixture
C. element D. compound

4. Solve problems :

A. What is the ratio of molecules in 1 mole of NH_3 and 1 mole of HNO_3.
(Ans. : 1:1)

B. Calculate number of moles of hydrogen in 0.448 litre of hydrogen gas at STP
(Ans. : 0.02 mol)

C. The mass of an atom of hydrogen is 1.008 u. What is the mass of 18 atoms of hydrogen. (18.144 u)

D. Calculate the number of atom in each of the following (Given : Atomic mass of I = 127 u).
a. 254 u of iodine (I)
b. 254 g of iodine (I)
(Ans. : 2 atoms, 1.2044×10^{24} atoms)

E. A student used a carbon pencil to write his homework. The mass of this was found to be 5 mg. With the help of this calculate.
a. The number of moles of carbon in his homework writing.
(Ans : 4.16×10^{-4})

b. The number of carbon atoms in 12 mg of his homework writting

(Ans : 6.022×10^{20})

F. Arjun purchased 250 g of glucose ($C_6H_{12}O_6$) for Rs 40. Find the cost of glucose per mole.

(Ans : Rs 28.8)

G. The natural isotopic abundance of ^{10}B is 19.60% and ^{11}B is 80.40 %. The exact isotopic masses are 10.13 and 11.009 respectively. Calculate the average atomic mass of boron

(Ans. :10.81)

H. Convert the following degree Celsius temperature to degree Fahrenheit.

a. 40 ^{0}C b. 30 ^{0}C

(Ans. : A. 104 ^{0}F, B. 86 ^{0}F)

I. Calculate the number of moles and molecules of acetic acid present in 22 g of it.

(Ans. : 0.3666 mol, 2.2076×10^{23} molecules)

J. 24 g of carbon reacts with some oxygen to make 88 grams of carbon dioxide. Find out how much oxygen must have been used.

(Ans. : 64.0)

K. Calculate number of atoms is each of the following. (Average atomic mass : N = 14 u, S = 32 u)

a. 0.4 mole of nitrogen

b. 1.6 g of sulfur

(Ans. : A. 2.4088×10^{23} , B. 3.011×10^{22} atom)

L. 2.0 g of a metal burnt in oxygen gave 3.2 g of its oxide. 1.42 g of the same metal heated in steam gave 2.27 of its oxide. Which law is verified by these data ?

M. In two moles of acetaldehyde (CH_3CHO) calculate the following

a. Number of moles of carbon

b. Number of moles of hydrogen

c. Number of moles of oxygen

d. Number of molecules of acetaldehyde

(Ans. : A. 4 mol, B. 8 mol, C. 2 mol, D. 12.044×10^{23} molecules)

N. Calculate the number of moles of magnesium oxide, MgO in i. 80 g and ii. 10 g of the compound. (Average atomic masses of Mg = 24 and O = 16)

(Ans. i. 2 mol ii. 0.25 mol)

O. What is volume of carbon dioxide, CO_2 occupying by i. 5 moles and ii. 0.5 mole of CO_2 gas measured at STP.

(Ans. i. 112 dm^3 ii. 11.2dm^3)

P. Calculate the mass of potassium chlorate required to liberate 6.72 dm^3 of oxygen at STP. Molar mass of $KClO_3$ is 122.5 g mol^{-1}.

(Ans. 24.5 g)

Q. Calculate the number of atoms of hydrogen present in 5.6 g of urea, $(NH_2)_2CO$. Also calculate the number of atoms of N, C and O.

(Ans. : No. of atoms of H = 2.24×10^{23}, N =1.124×10^{23} and C = 0.562×10^{23}, O = 0.562×10^{23})

R. Calculate the mass of sulfur dioxide produced by burning 16 g of sulfur in excess of oxygen in contact process. (Average atomic mass : S = 32 u, O = 16 u)

(Ans. 32 g)

5. Explain

A. The need of the term average atomic mass.

B. Molar mass.

C. Mole concept.

D. Formula mass with an example.

E. Molar volume of gas.

F. Types of matter (on the basis of chemical composition).



Activity :

Collect information of various scientists and prepare charts of their contribution in chemistry.

2. Introduction to analytical chemistry

2.1 Intrduction : Analytical chemistry facilitates investigation of chemical composition of substances. It uses the instruments and methods to separate, identify and quantify the matter under study. The analysis thus provides chemical or physical information about a sample. Analysis may be qualitative or quantitative. Qualitative analysis is concerned with the detection of the presence or absence of elements in compounds and mixture of compounds. Quantitative analysis deals with the determination of the relative proportions of elements in compounds and mixture compounds.



Remember

The branch of chemistry which deals with the study of seperation, identification, qualitative and quantitative determination of the compositions of different substances, is called analytical chemistry.

Importance of analytical chemistry :

The course of analytical chemistry extends the knowledge acquired by the students in studying general, inorganic and organic chemistry. Chemical analysis is one of the most important methods of monitoring the composition of raw materials, intermediates and finished products, and also the composition of air in streets and premises of industrial plants. In agriculture, chemical analysis is used to determine the compostion of soils and fertilizers; in medicine, to determine the composition of medicinal preparations.Analytical chemistry has applications in forensic science, engineering and industry. Industrial process as a whole and the production of new kinds of materials are closely associated with analytical chemistry. Analytical chemistry consists of classical, wet chemical methods and modern instumental methods.

2.2 Analysis : Analysis is carried out on a small sample of the material to be tested, and not on the entire bulk. When the amount of a solid or liquid sample is a few grams, the analysis is called semi-microanalysis. It is of two types : **qualitative and quantitative**. Classical qualitative analysis methods include separations such as precipitation, extraction and distillation. Identification may be based on differences in colour, odour, melting point, boiling point, and reactivity. Classical quantitative methods consist of volumetric analysis, gravimetric analysis, etc.

2.2.1 Chemical methods of qualitative analysis : Chemical analysis of a sample is carried out mainly in two stages : by the dry method in which the sample under test is not dissolved and by the wet method in which the sample under test is first dissolved and then analyzed to determine its composition. The dry method is usually used as preliminary tests in the qualititative analysis.

The semi-micro qualitative analysis is carried out using apparatus such as : test tubes, beakers, evaporating dish, crucible, spot plate, watch glass, wire guaze, water bath, burner, blow pipe, pair of tongues, centrifuge, etc.

The qualitative analysis of **organic** and **inorganic** compounds involves **different** types of **tests**. The majority of organic compounds are composed of a relatively small number of elements. The most important ones are : carbon, hydrogen, oxygen, nitrogen, sulphur, halogen, phosphorous. Elementary qualitative analysis is concerned with the detection of the presence of these elements. The identification of an organic compound involves tests such as detection of functional group, determinition of melting/ boiling point, etc. The qualitative analysis of simple inorganic compounds involves detection and confirmation of cationic

and anionic species (basic and acidic radical) in them.

2.2.2 Chemical methods of quantitative analysis : Quantitative analysis of organic compounds involves methods such as (i) determination of percentage constituent element, (ii) concentrations of a known compound in the given sample, etc. Quantitative analysis of simple inorganic compounds involves methods based on (i) decomposition reaction (gravimetric analysis), and (ii) the progress of reaction between two solutions till its completion (titrametric or volumetric analysis), etc. The quantitative analytical methods involve measurement of quantities such as mass and volume, by means of some equipment/ apparatus such as weighing machine, burette.

2.3 Mathematical operation and error analysis : The accuracy of measurement is of a great concern in analytical chemistry. Also there can be intrinsic errors in the analytical measurement. The numerical data, obtained experimentally, are treated mathematically to reach some quantitative conclusion. Therefore, an anlytical chemist has to know how to report the quantitative analytical data, indicating the extent of the accuracy of measurement, perform the mathematical operation and properly express the quatitative error in the result. In the following subsection we will consider these aspects related to measurments and calculation.

2.3.2 Scientific notation (exponential notation) : A chemist has to deal with numbers as large as 602,200,000,000, 000, 000, 000, 000 for the molecules of 2 g of hydrogen gas or as small as 0.00000000000000000000000166g. that is, mass of a H atom. To avoid the writing of so many zeros in mathematical operations, scientific notations i.e. exponential notations are used. Here, any number can be represented into a form $N \times 10^n$ where 'n' is an exponent having positive or negative values and N can vary $1 \leqslant N < 10$. Thus, we can write the above values as 6.022×10^{23} and 1.66×10^{-24} g. The number 123.546 becomes 1.23546×10^2, in scientific notation. Note that while writing it, we have moved the decimal to the left by two places and same is the exponent (2) of 10 in the scientific notation. Similarly, 0.00015 can be written as 1.5×10^{-4}.

Problem 2.1 : For adding 5.55×10^4 and 6.95×10^3, first the exponent is made equal. Thus
$5.55 \times 10^4 + 0.695 \times 10^4$. Then these numbers can be added as follows :
$(5.55 + 0.695) \times 10^4 = 6.245 \times 10^4$

Problem 2.2 : The subtraction of two numbers can be done as shown below:
$3.5 \times 10^{-2} - 5.8 \times 10^{-3}$
$= (3.5 \times 10^{-2}) - (0.58 \times 10^{-2})$
$= (3.5 - 0.58) \times 10^{-2}$
$= 2.92 \times 10^{-2}$

Here the decimal has to be moved four places to the right and (-4) is the exponent in the scientific notation. Now let us perform mathematical operations on numbers expressed in scientific notation.

Problem 2.3 : $(5.6 \times 10^5) \times (6.9 \times 10^8)$
$= (5.6 \times 6.9) (10^{5+8})$
$= (5.6 \times 6.9) \times 10^{13}$
$= 38.64 \times 10^{13}$
$= 3.864 \times 10^{14}$

Problem 2.4 : $(9.8 \times 10^{-2}) \times (2.5 \times 10^{-6})$
$= (9.8 \times 2.5) (10^{-2 + (-6)})$
$= (9.8 \times 2.5) \times (10^{-2-6})$
$= 24.50 \times 10^{-8}$
$= 2.45 \times 10^{-7}$

Addition and subtraction : To perform addition operation, first the numbers are written in such a way that they have the same exponent. The coefficients are then added. (Problems 2.1 and 2.2)

Multiplication : The rule for the multiplication of exponential numbers can be well explained from the solved problems 2.3 and 2.4.

2.3.2 Precision and accuracy of measurement

Aim of any measurement is to get the actual value called true value or accepted value of a quantity. Nearness of the measured value to the true value is called the accuracy of measurement. Larger the accuracy smaller the error. Accuracy depends upon the sensitivity or least count (the smallest quantity that can be measured) of the measuring quuipment. consider, for example, a burette reading of 10.2 mL. For all the three situations in the Fig. 2.1 the reading would be noted as 10.2 mL It means that there is an uncertainty about the digit appearing after the decimal point in the reading 10.2 mL. This is because the least count of the burette is 0.1 mL. The meaning of the reading 10.2 mL is that the true value of the reading lies between 10.1 mL and 10.3 mL. This is indicated by writing 10.2 ± 0.1 mL. Here, the burette reading has an error of ± 0.1mL (Fig. 2.1).

Errors may be expressed as **absolute** or **relative error**.

Absolute error = Observed value - True value

Relative error is generally a more useful quantity than absolute error. Relative error is the ratio of an absolute error to the true value. It is expressed as a percentage.

$$\text{Relative error} = \frac{\text{Absolute error}}{\text{True value}} \times 100\ \%$$

There can be error in a measurement due to a number of reasons including inefficiency of the person doing measurement.

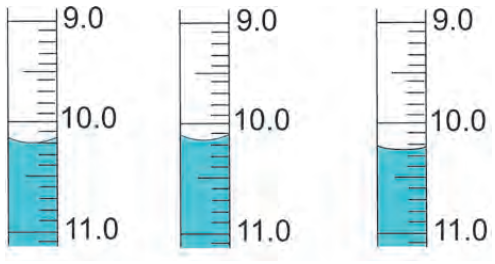


Fig. 2.1 : Three possibilities of a burette reading 10.2 mL

Multiple readings of the same quantity are noted to minimize the error. If the readings match closely, they are said to have high precision. High percision implies reproducibility of the readings. **High precision is a prerequisite for high accuracy**. Precision is expressed in terms of deviation. An absolute deviation is the modulus of the difference between an observed value and the arithmetic mean for the set of several measurements made in the same way. It is a measure of absolute error in the repeated observation.

Absolute deviation = |Observed value - Mean|

Arithmetic mean of all the absolute deviations is called the mean absolute deviation in the measurements. The ratio of mean absolute deviation to its arithmentic means is called relative deviation.

$$\text{Relative deviation} = \frac{\text{Mean absolute deviation}}{\text{Mean}} \times 100\ \%$$

Problem 2.5 : In laboratory experiment, 10 g potassium chlorate sample on decomposition gives following data ; The sample contains 3.8 g of oxygen and the actual mass of oxygen in the quantity of potassium chlorate is 3.92 g. Calculate absolute error and relative error.

Solution : The observed is 3.8 g and accepted value is 3.92 g

Absolute error = Observed value - True value

$= 3.8 - 3.92 = -0.12$ g

The negative sign indicates that your experimental result is lower than the true value.

$$\text{The relative error} = \frac{\text{Absolute error}}{\text{True value}} \times 100\%$$

$$= \frac{-0.12}{3.92} \times 100\%$$

$$= -3.06\ \%$$

Problem 2.6 : The three identical samples of potassium chlorate are decomposed. The mass of oxygen is determined to be 3.87 g, 3.95 g and 3.89 g for the set. Calculate absolute deviation and relative deviation.

Solution :

$$\text{mean} = \frac{3.87 + 3.95 + 3.89}{3} = 3.90$$

Average deviation of a set of measurements		
Sample	**Mass of oxygen**	**Deviation**
1	3.87g	0.03g
2	3.95g	0.05g
3	3.89g	0.01g
Mean absolute deviation		0.03

Absolute deviation

$= |\text{Observed value} - \text{Mean}|$

$\therefore$ Mean absolute deviation = ± 0.03 g.

The relative deviation,

$$= \frac{\text{Mean absolute deviation}}{\text{Mean}} \times 100\ \%$$

$$= \frac{0.03}{3.9} \times 100\% = 0.8\%$$

2.3.3 Significant Figures : Uncertainty in measured value leads to uncertainty in calculated result. Uncertainty in a value is indicated by mentioning the number of significant figures in that value. Consider, the column reading 10.2 ± 0.1 mL recorded on a burette having the least count of 0.1 mL. Here it is said that the last digit '2' in the reading is uncertain, its uncertainty is ±0.1 mL. On the other hand, the figure '10' is certain. The significant figures in a measurement or result are the number of digits known with certainty plus one uncertain digit. In a scientific experiment a result is obtained by doing calculation in which values of a number of quantities measured with equipment of different least counts are used.

Following rules are to be followed during such calculation.

2.3.4 Rules for deciding significant figures :

1. All non zero digits are significant; e. g. 127.34 g contains five significant figures which are 1, 2, 7, 3 and 4.
2. All zeros between two non zero digits are significant e. g. 120.007 m contains six significant figures.
3. Zeroes on the left of the first non zero digit are not significant. Such a zero indicates the position of the decimal point. For example, 0.025 has two significant figures, 0.005 has one significant figure.
4. Zeroes at the end of a number are significant if they are on the right side of the decimal point. Terminal zeros are not significant if there is no decimal point. (This is beacause the least count of an instrument contains decimal point) For example 0.400 g has three singnificant figures. The measurements here indicates that it is made on a weighing machine having least count of 0.001 g. Significant figures are also indicated in scientific notation by means of decimal point. For example, the measurment 400 g has one significant figure. The measurement 4.0×10^2 g has two significant figures, wheras the measurment 4.00×10^2 g has three significant figures. The zeros after the decimal points in these cases indicates that the least counts of the weighing machines are 1 g, 0.1 g and 0.01 g, respectively.
5. In numbers written is scientific notation, all digits are significant. For example, 2.035×10^2 has four significant figures, and 3.25×10^{-5} has three significant figures.

Problem 2.7 : How many significant figures are present in the following measurements ?

a. 4.065 m b. 0.32 g c. 57.98 cm^3
d. 0.02 s e. 4.0 x 10^{-4} km
f. 604.0820 kg g. 307.100 x 10^{-5} cm

Ans. : a. 4 b. 2 c. 4 d. 1
e. 2 f. 7 g. 6

In general, a quantity measured with an instrument of smaller least count will have more significant figures and will be more accurate than when measured with an instrument of larger least count.

2.3.5 Calculations with significant figures :

When performing calculations with measured quantities the rule is that the accuracy of the final result is limited to the accuracy of the least accurate measurement. In other words, the final result can not be more accurate than the least accurate number involved in the calculation.

Rounding off : The final result of a calculation often contains figures that are not significant. When this occurs the final result is rounded off. The following rules are used to round off a number to the required number of significant figures :

If the digit following the last digit to be kept is less than five, the last digit is left unchanged.

e.g. 46.32 rounded off to two significant figures is 46.

If the digit following the last digit to be kept is five or more, the last digit to be kept is increased by one. e.g. 52.87 rounded to three significant figures is 52.9.

Problem 2.8 : Round off each of the following to the number of significant digits indicated :
a. 1.223 to two digits b. 12.56 to three digits c. 122.17 to four digits d. 231.5 to three digits.
Ans. : i. 1. 2; the third digit is less than 5, so we drop it all the others to its right.
ii. 12.6 ; the fourth digit is greater than 5, so we drop it and add 1 to the third digit.
iii. 122.2 ; the fifth digit is greater than 5, so we do it and add 1 to the fourth digit.
iv. 232; the fourth digit is 5, so we drop it and add 1 to the third digit.

2.4 Determination of molecular formula :

Molecular formula of a compound is the formula which indicates the actual number of atoms of the constituent elements in a molecule. It can be obtained from the experimentally determined values of percent elemental composition and molar mass of that compound.

2.4.1 Percent composition and empirical formula : Compounds are formed by chemical combination of different elements. Quantitative determination of the constituent element by suitable methods provides the percent elemental composition of a compound. If the percent total is not 100, the difference is considered as percent oxygen. From the the per cent composition, the ratio of the atoms of the constituent elements in the molecule is calculated. **The simplest ratio of atoms of the constituent elements in a molecule is called the empirical formula of that compound**. Molecular formula can be obtained from the empirical formula if the molar mass is known. The molar mass of the substance under examination is determined by some convenient method. The following example illustrates this sequence.

Problem 2.9 : A compound contains 4.07 % hydrogen, 24.27% carbon and 71.65 % chlorine by mass. Its molar mass is 98.96 g. What is its empirical formula ? Atomic masses of hydrogen, carbon and chlorine are 1.008, 12.000 and 35.453 u, respectively
Solution :
Step I : Check whether the sum of all the percentages is 100.
$4.07 + 24.27 + 71.65 = 99.99 \approx 100$
Therefore no need to consider presence of oxygen atom in the molecule.
Step II : Conversion of mass percent to grams. Since we are having mass percent, it is convenient to use 100 g of the compound as the starting material. Thus in the 100 g sample of the above compound, 4.07 g hydrogen 24.27 g carbon and 71.65 g chlorine is present..... Contd on next page

Step III : Convert into number/of moles of each element. Divide the masses obtained above by respective atomic masses of various elements.

$$\text{Moles of hydrogen} = \frac{4.07\ g}{1.008\ g} = 4.04$$

$$\text{Moles of carbons} = \frac{24.27\ g}{12.01\ g} = 2.0225$$

$$\text{Moles of chlorine} = \frac{71.65\ g}{35.453\ g} = 2.021$$

Steps IV : Divide the mole values obtained above by the smallest value among them.
Since 2.021 is smallest value, division by it gives a ratio of 2:1:1 for H:C:Cl.
In case the ratio are not whole numbers, then they may be converted into whole number by multiplying by the suitable coefficient.

Step V : Write empirical formula by mentioning the numbers after writing the symbols of respective elements. CH_2Cl is thus, the empirical formula of the above compound.

Step VI : Writing molecular formula

a. Determine empirical formula mass : Add the atomic masses of various atoms present in the empirical formula.
For CH_2Cl, empirical formula mass is
$12.01 + 2 \times 1.008 + 35.453$
$= 49.48\ g$

b. Divide molar mass by empirical formula mass

$$\therefore \frac{\text{Molar mass}}{\text{Empirical formula mass}} = \frac{98.96\ g}{49.48\ g}$$

$\therefore r = 2$

c. multiply empirical formula by r obtained above to get the molecular formula.
Molecular formula = r x empirical formula
molecular formula is 2 x CH_2Cl i.e. $C_2H_4Cl_2$.

Problem 2.10 : A compound with molar mass 159 was found to contain 39.62 % copper and 20.13 % sulfur. Suggest molecular formula for the compound (Atomic masses: Cu = 63, S = 32 and O = 16).

Solution :
% copper + % sulfur = 39.62 + 20.13
= 59.75
This is less than 100 % Hence compound contains adequate oxygen so that the total percentage of elements is 100%.
Hence % of oxygen = 100 - 59.75 = 40.25%

$$\text{Moles of Cu} = \frac{\%\ \text{of Cu}}{\text{Atomic mass of Cu}}$$

$$= \frac{39.62}{63} = 0.629$$

$$\text{Moles of S} = \frac{\%\ \text{of S}}{\text{Atomic mass of S}}$$

$$= \frac{20.13}{32} = 0.629$$

$$\text{Moles of O} = \frac{\%\ \text{of O}}{\text{Atomic mass of O}} = \frac{40.25}{16} = 2.516$$

Hence the ratio of number of moles of Cu:S:O is

$$\frac{0.629}{0.629} = 1 \quad \frac{0.629}{0.629} = 1 \text{ and } \frac{2.516}{0.629} = 4$$

Hence empirical formula is $CuSO_4$
Empirical formula mass
$= 63 + 32 + 16 \times 4 = 159$
Molar mass = Empirical mass (Since Molar mass = Molecular mass)
$\therefore$ Molecular formula = Empirical formula
= $CuSO_4$

2.5 Chemical reactions and stoichiometric calculations

Calculation based on a balanced chemical equations are known as stoichiometric calculations.

Balanced chemical equation is symbolic representation of a chemical reaction. It supplies the following information which is useful in solving problems based on chemical equations

i. It indicates the number of moles of the reactants involved in a chemical reaction and the number of moles of the products formed.

ii. It indicates the relative masses of the reactants and products linked with a chemical change, and

iii. it indicates the relationship between the volume/s of the gaseous reactants and products, at STP.

2.5.1 Stoichiometric problems

Generally problems based on stoichiometry are of the following types :

a. Problems based on mass-mass relationship;

b. Problems based on mass-volume relationship and

c. Problems based on volume-volume relationship.

Steps involved in problems based on stoichiometric calculations :

1. Write down the balanced chemical equation representing the chemical reaction.
2. Write the number of moles and the relative masses or volumes of the reactants and products below the respective formulae.
3. Relative masses or volumes should be calculated from the respective formula mass referring to the condition of STP.
4. Apply the unitary method to calculate the unknown factor/s as required by the problem.

Problem 2.11 : Calculate the mass of carbon dioxide and water formed on complete combustion of 24 g of methane gas. (Atomic masses, C = 12 u, H = 1 u, O = 16 u)

Solution : The balanced chemical equation is,

$CH_4(g) + 2\ O_2(g) \rightarrow CO_2(g) + 2\ H_2O(g)$

$(12 + 4)\quad 2 \times (16 \times 2)\quad 12 + (16 \times 2)\quad 2 \times (2 + 16)$

$= 16\ g \qquad = 64\ g \qquad = 44\ g \qquad = 36\ g$

Hence, 16 g of CH_4 on complete combustion will produce 44 g of CO_2

$\therefore$ 24 g of $CH_4 = \frac{24}{16} \times 44 = 66$ g of CO_2

Similarly, 16 g of CH_4 will produce 36 g of water.

24 g of $CH_4 \equiv \frac{24}{16} \times 36 = 54$ g water.

Problem 2.12 : How much CaO will be produced by decomposition of 5g $CaCO_3$?

Solution : Calcium carbonate decomposes according to the balanced equation,

$CaCO_3 \xrightarrow{\Delta} CaO + CO_2\uparrow$

$40 + 12 + 3 \times 16 \qquad 40 + 16 \qquad 12 + 2 \times 16$

$= 100$ parts $\qquad = 56$ parts $\quad = 44$ parts

So, 100 g of $CaCO_3$ produces 56 g of CaO

$\therefore$ 5 g of $CaCO_3$ will produce

$= \frac{56\ \cancel{g}}{100\ \cancel{g}} \times 5g \quad = 2.8$ g of $CaCO_3$

Problem 2.13 : How many litres of oxygen at STP are required to burn completely 2.2 g of propane, C_3H_8 ?

Solution : The balanced chemical equation for the combustion of propane is,

$C_3H_8 + 5\ O_2 \rightarrow 3\ CO_2 + 4\ H_2O$

$3 \times 12 + 8 \times 1 \qquad 5 \times 22.4\ L$

$(44\ g) \qquad (112\ L)$

(Where 1 mol of ideal gas occupies 22.4 L of volume)

Thus 44 g of propane requires 112 litres of oxygen for complete combustion

$\therefore$ 2.2 g of propane will require

$\frac{112}{44} \times 2.2 = 5.6$ litres of O_2 at STP for complete combustion.

Problem 2.14 : A piece of zinc weighing 0.635 g when treated with excess of dilute H_2SO_4 liberated 200 cm^3 of hydrogen at STP. Calculate the percentage purity of the zinc sample.

Solution : The relevant balanced chemical equation is,

$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$

It indicates that 22.4 L of hydrogen at STP = 65 g of Zn.

(where Atomic mass of Zn = 65 u)

$\therefore$ 0.200 L of hydrogen at STP

$= \frac{65g}{22.4\ L} \times 0.200\ L \quad = 0.58$ g

$\therefore$ percentage purity of Zn $= \frac{0.58}{0.635} \times 100$

$= 91.33\ \%$

2.6 Limiting reagent

When a chemist carries out a reaction, the reactants are not usually present in exact stoichiometric amounts, that is, in the proportions indicated by the balanced equation. Because the goal of a reaction is to produce the maximum quantity of a useful compound from the starting materials, frequently, a large excess of one reactant is supplied to ensure that the more expensive reactant is completely converted into the desired product. The reactant which is present in lesser amount gets consumed after some time and subsequently, no further reaction takes place, whatever be the amount left of the other reactant present. Hence, the reactant which gets consumed, limits the amount of product formed and is therefore, called the **limiting reagent**.

Consider the formation of nitrogen dioxide (NO_2) from nitric axide (NO) and oxygen

$$2NO\,(g) + O_2\,(g) \longrightarrow 2NO_2\,(g)$$

Suppose initially we have 8 moles of NO and 7 moles of O_2. One way to determine which of the two reactants in the limiting reagent is to calculate the number of moles NO_2 obtainable from the given initial quantities of NO and O_2.

From the preceding definition, we see that the limiting reagent will yield the smaller amount of the product. Starting with 8 moles of NO, we find the number of NO_2 produced is

$$8 \text{ mol NO} \times \frac{2 \text{ mol } NO_2}{2 \text{ mol NO}} = 8 \text{ mol } NO_2$$

and starting with 7 moles of O_2, the number of moles NO_2 formed is

$$7 \text{ mol } O_2 \times \frac{2 \text{ mol } NO_2}{1 \text{ mol } O_2} = 14 \text{ mol } NO_2$$

Because 8 moles NO result in a smaller amount of NO_2, NO must be the limiting reagent, and O_2 is the excess reagent, before reaction has started.

Problem 2.15 : Urea [$(NH_2)_2CO$] is prepared by reacting ammonia with carbon dioxide.

$$2NH_3(g) + CO_2(g) \rightarrow (NH_2)_2CO\,(aq) + H_2O(l)$$

In one process, 637.2 g of NH_3 are treated with 1142 g of CO_2. (a) Which of the two reactants is the limiting reagent ? (b) Calculate the mass of $(NH_2)_2CO$ formed. (c) How much excess reagent (in grams) is left at the end of the reaction ?

Solution : (a) We carry out two separate calculations. First : If 637.2 g of NH_3 reacts completely, calculate the number of moles of $(NH_2)_2CO$, that could be produced, by the following relation.

mass of NH_3 → moles of NH_3 → moles of $(NH_4)_2CO$

$$\text{moles of } (NH_2)_2CO = 637.2 \text{ g } NH_3 \times \frac{1 \text{ mol } NH_3}{17.03 \text{ g } NH_3} \times \frac{1 \text{ mol } (NH_4)_2CO}{2 \text{ mol } NH_3}$$

$$= 18.71 \text{ moles } (NH_2)_2CO$$

Second : The relation from 1142 g of CO_2:

mass of CO_2 → moles of CO_2 → moles of $(NH_2)_2CO$

The number of moles of $(NH_2)_2CO$ that could be produced if all the CO_2 reacted :

$$\text{moles of } (NH_2)_2CO = 1142 \text{ g } CO_2 \times \frac{1 \text{ mol } CO_2}{44.01 \text{ g } CO_2} \times \frac{1 \text{ mol } (NH_2)_2CO}{1 \text{ mol } CO_2}$$

$$= 25.95 \text{ mol } (NH_2)_2CO$$

It follows, therefore, that NH_3 must be the limiting reagent because it produces (a) smaller amount of $(NH_2)_2CO$) (b) The molar mass of $(NH_2)_2CO$ is 60.06 g. We use this as a conversion factor to convert from moles of $(NH_2)_2CO$ to grams of $(NH_2)_2CO$.

$$\text{mass of } (NH_2)_2CO = 18.71 \text{ mol } (NH_2)_2CO \times \frac{60.06 \text{ g } (NH_2)_2CO}{1 \text{ mol}}$$

$$\therefore \quad = 1124 \text{ g } (NH_2)_2CO$$

(c) Starting with 18.71 moles of $(NH_2)_2CO$, we can determine the mass of CO_2 that reacted using the mole ratio from the balanced equation and the molar mass of CO_2

.....Contd. on next page

The conversion steps are
moles of $(NH_2)_2CO$ ⟶ moles of CO_2 ⟶ grams of CO_2
So that,
mass of CO_2 reacted
$= 18.71 \text{ mol } (NH_2)_2CO \times \frac{1 \text{ mol } CO_2}{1 \text{ mol } (NH_2)_2CO} \times \frac{44.01 \text{ g } CO_2}{1 \text{ mol } CO_2}$

$= 823.4$ g
The amount of CO_2 remaining (in excess) is the difference between the initial amount (1142 g) and the amount reacted (823.4 g):
mass of CO_2 remaining = 1142 g - 823.4 g
= 318.6 g ≈ 319 g

2.7 Concentration of solution : A majority of reactions in the laboratory are carried out in solutions. Therefore, it is important to understand how the amount of substance is expressed when it is present in the form of a solution. The concerntration of a solution or the amount of substance present in given volume of a solution can be expressed in any of the following ways :

1. Mass percent or weight percent (w/w %)
2. Mole fraction
3. Molarity (M)
4. Molality (m)

2.7.1 Mass percent : It is obtained by using following relation:

$$\text{Mass percent} = \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100\ \%$$

Problem 2.16 : A solution is prepared by adding 2 g of a substance A to 18 g of water. Calculate the mass percent of the solute.

Solution : Mass percent of

$$A = \frac{\text{Mass of A}}{\text{Mass of solution}} \times 100$$

$$= \frac{2 \text{ g}}{2 \text{ g of A} + 18 \text{ g of water}} \times 100$$

$$= \frac{2 \text{ g}}{20 \text{ g}} \times 100 = 10\ \%$$

2.7.2 Mole fraction

It is the ratio of number of moles of a particular component of a solution to the total number of moles of the solution. If a substance 'A' dissolves in substance 'B' and their number of moles are n_A and n_B , repsectively, then the mole fraction of A and B are given as :

$$\text{Mole fraction of A} = \frac{\text{No. of moles of A}}{\text{No. of moles of solution}}$$

$$\therefore \text{ Mole fraction of A } = \frac{n_A}{n_A + n_B}$$

$$\text{Mole fraction of B} = \frac{\text{No. of moles of B}}{\text{No. of moles of solution}}$$

$$= \frac{n_B}{n_A + n_B}$$

2.7.3 Molarity

It is the most wideley used unit and is denoted by M. It is defined as the number of moles of the solute present in 1 litre of the solution . Thus,

$$\text{Molarity (M)} = \frac{\text{No. of moles of solute}}{\text{Volume of solution in litres}}$$

Problem 2.17 : Calculate the molarity of NaOH in the solution prepared by dissolving its 4 g in enough water to form 250 mL of the solution.

Solution :

Since molarity (M) =

$$\frac{\text{No. of moles of solute}}{\text{Volume of solution in litres}}$$

$$\therefore \text{ M} = \frac{\text{Mass of NaOH / Molar mass of NaOH}}{0.250 \text{ L}}$$

$$\therefore \text{ M} = \frac{4 \text{ g} / 40 \text{ g}}{0.250 \text{ L}}$$

$$\therefore \text{ M} = \frac{0.1 \text{ mol}}{0.250 \text{ L}}$$

$$\therefore \text{ M} = 0.4 \text{ mol L}^{-1} = 0.4 \text{ M}$$

Note that molarity of a solution depends upon temperature because volume of a solution is temperature dependent.

2.7.4 Molality

It is defined as the number of moles of solute present in 1 kg of solvent. It is denoted by m.

$$\text{Molality (m)} = \frac{\text{No. of moles of solute}}{\text{Mass of solvent in kg}}$$

Note that molality of a solution does not change with temperature since mass remains unaffected with temperature. Often in chemistry laboratory, a solution of desired concentration is prepared by diluting a solution of known higher concentration. The solution of higher concentration is also known as stock solution.

Problem 2.18 : The density of 3M solution of NaCl is 1.25 g mL^{-1} Calculate molality of the solution.

Solution : Molarity = 3 mol L^{-1}

Mass of NaCl in 1 L solution $= 3 \times 58.5$

$\therefore$ $= 175.5$ g

Mass of 1L solution = $1000 \times 1.25 = 1250$ g

($\because$ density = 1.25 g mL^{-1})

Mass of water in solution = 1250 - 175.5

$\therefore$ $= 1074.5$ g

$$\text{Molality} = \frac{\text{No. of moles of solute}}{\text{Mass of solvent in kg}}$$

$$= \frac{3 \text{ mol}}{1.0745 \text{ kg}} = 2.79 \text{ m}$$

2.8 Use of graph in analysis : Analytical chemistry also involves deducing some relation, if any, between two or more properties of matter under study. One of the classic example in the relation between temperature and volume of a given amount of gas. A set of experimentally measured values of volume and temperature of a definite mass of a gas upon plotting on a graph paper appeared as in the figure (Fig. 2.2 (a)). When the points are directly connected, a zig zag pattern results (Fig. 2.2 (b)). From this pattern no meaningful result can be deduced. A zig zag pattern results due to many types of errors that incur in many measurements involved an experiment. Figure 2.2 (c) shows a smooth curve which may be called an average curve passing through these points. In the above example it happens to be straight line and the inference is that $V \propto T$.

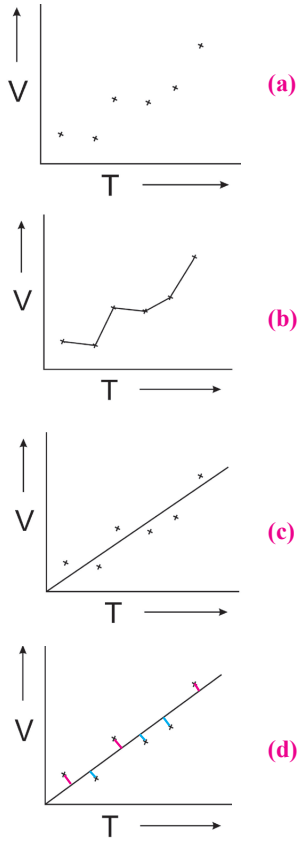


Fig. 2.2 : Drawing an average curve through the points on graph

While fitting it to a smooth curve, care is taken that the plotted points are evenly distributed about it. Mathematically 'even distribution' is understood as follows :

From each point draw a perpendicular to the curve. The perpendicular represents deviation of each point from the curve (Fig 2.2 (d)). The positive deviations are shown in red and negative deviations are shown in blue. Take sum of all the red perpendiculars and all the blue perpendiculars separately. If the two sums are equal (or nearly equal) the curve drawn shows the experimental points in the best possible representation.

Exercises

1. Choose correct option

A. The branch of chemistry which deals with study of separation, identification, and quantitaive determination of the composition of different substances is called as .....
   a. Physical chemistry
   b. Inorganic chemistry
   c. Organic chemistry
   d. Analytical chemistry

B. Which one of the following property of matter is Not quantitative in nature ?
   a. Mass b. Length
   c. Colour d. Volume

C. SI unit of mass is .....
   a. kg b. mol
   c. pound d. m^3

D. The number of significant figures in 1.50×10^4 g is .....
   a. 2 b. 3
   c. 4 d. 6

E. In Avogadro's constant 6.022×10^{23} mol^{-1}, the number of significant figures is .....
   a. 3 b. 4
   c. 5 d. 6

F. By decomposition of 25 g of $CaCO_3$, the amount of CaO produced will be .....
   a. 2.8 g b. 8.4 g
   c. 14.0 g d. 28.0 g

G. How many grams of water will be produced by complete combustion of 12 g of methane gas
   a. 16 b. 27 c. 36 d. 56

H. Two elements A (At. mass 75) and B (At. mass 16) combine to give a compound having 75.8 % of A. The formula of the compound is
   a. AB b. A_2B
   c. AB_2 d. A_2B_3

I. The hydrocarbon contains 79.87 % carbon and 20.13 % of hydrogen. What is its empirical formula ?
   a. CH b. CH_2
   c. CH_3 d. C_2H_5

J. How many grams of oxygen will be required to react completely with 27 g of Al? (Atomic mass : Al = 27, O = 16)
   a. 8 b. 16
   c. 24 d. 32

K. In $CuSO_4.5H_2O$ the percentage of water is .....
   (Cu = 63.5, S = 32, O = 16, H = 1)
   a. 10 % b. 36 %
   c. 60 % d. 72 %

L. When two properties of a system are mathematically related to each other, the relation can be deduced by
   a. Working out mean deviation
   b. Plotting a graph
   c. Calculating relative error
   d. all the above three

2. Answer the following questions

A. Define : Least count

B. What do you mean by significant figures? State the rules for deciding significant figures.

C. Distinguish between accuracy and precision.

D. Explain the terms percentage composition, empirical formula and molecular formula.

E. What is a limiting reagent ? Explain.

F. What do you mean by SI units ? What is the SI unit of mass ?

G. Explain the following terms
   (a) Mole fraction
   (b) Molarity
   (c) Molality

H. Define : Stoichiometry

I. Why there is a need of rounding off figures during calculation ?

J. Why does molarity of a solution depend upon temperature ?

M. Define Analytical chemistry. Why is accurate measurement crucial in science?

3. Solve the following questions

A. How many significant figures are in each of the following quantities ?
   a. 45.26 ft b. 0.109 in
   c. 0.00025 kg d. 2.3659×10^{-8} cm
   e. 52.0 cm^3 f. 0.00020 kg
   g. 8.50×10^4 mm h. 300.0 cg

B. Round off each of the following quantities to two significant figures :
   a. 25.55 mL b. 0.00254 m
   c. 1.491×10^5 mg d. 199 g

C. Round off each of the following quantities to three significant figures :
   a. 1.43 cm^3 b. 458×10^2 cm
   c. 643 cm^2 d. 0.039 m
   e. 6.398×10^{-3} km
   f. 0.0179 g g. 79,000 m
   h. 42,150 i. 649.85;
   j. 23,642,000 mm
   k. 0.0041962 kg

D. Express the following sum to appropriate number of significant figures :
   a. 2.3×10^3 mL + 4.22×10^4 mL + 9.04×10^3 mL + 8.71×10^5 mL;
   b. 319.5 g - 20460 g - 0.0639 g - 45.642 g - 4.173 g

4. Solve the following problems

A. Express the following quantities in exponential terms.
   a. 0.0003498 b. 235.4678
   c. 70000.0 d. 1569.00

B. Give the number of significant figures in each of the following
   a. 1.230×10^4 b. 0.002030
   c. 1.23×10^4 d. 1.89×10^{-4}

C. Express the quantities in above (B) with or without exponents as the case may be.

D. Find out the molar masses of the following compounds :
   a. Copper sulphate crystal ($CuSO_4.5H_2O$)
   (Ans.: 249.5 g/mol)
   b. Sodium carbonate, decahydrate ($Na_2CO_3.10H_2O$)
   (Ans.: 286 g/mol)
   c. Mohr's salt [$FeSO_4(NH_4)_2SO_4.6H_2O$]
   (Ans.: 392 g/mol)
   (At. mass : Cu = 63.5; S = 32; O = 16; H = 1; Na = 23; C = 12; Fe = 56; N = 14)

E. Work out the percentage composition of constituents elements in the following compounds :
   a. Lead phosphate [$Pb_3(PO_4)_2$],
   b. Potassium dichromate ($K_2Cr_2O_7$),
   c. Macrocosmic salt - Sodium ammonium hydrogen phosphate, tetrahydrate ($NaNH_4HPO_4.4H_2O$)
   (At. mass : Pb = 207; P = 31; O = 16; K = 39; Cr = 52; Na = 23; N = 14)

F. Find the percentage composition of constituent green vitriol crystals ($FeSO_4.7H_2O$). Also find out the mass of iron and the water of crystallisation in 4.54 kg of the crystals. (At. mass : Fe = 56; S = 32; O = 16)
   (Ans.: mass of Fe = 0.915 kg, mass of $7H_2O$ = 2.058 kg)

G. The red colour of blood is due to a compound called "haemoglobin". It contains 0.335 % of iron. Four atoms of iron are present in one molecule of haemoglobin. What is its molecular weight ? (At. mass : Fe 55.84)
   (Ans.: 66674.6 g/mol)

H. A substance, on analysis, gave the following percent composition: Na = 43.4 %, C = 11.3 % and O = 45.3 %. Calculate the empirical formula. (At. mass Na = 23 u, C = 12 u, O = 16 u).
   (Ans.: Na_2CO_3)

I. Assuming the atomic weight of a metal M to be 56, find the empirical formula of its oxide containing 70.0% of M.
   (Ans.: M_2O_3)

J. 1.00 g of a hydrated salt contains 0.2014 g of iron, 0.1153 g of sulfur, 0.2301 g of oxygen and

0.4532 g of water of crystallisation. Find the empirical formula. (At. wt. : Fe = 56; S = 32; O = 16)

(Ans.: $FeSO_4$)

K. An organic compound containing oxygen, carbon, hydrogen and nitrogen contains 20 % carbon, 6.7 % hydrogen and 46.67 % nitrogen. Its molecular mass was found to be 60. Find the molecular formula of the compound.

(Ans.: CH_4N_2O)

L. A compound on analysis gave the following percentage composition by mass : H = 9.09; O = 36.36; C = 54.55. Mol mass of compound is 88. Find its molecular formula.

M. Carbohydrates are compounds containing only carbon, hydrogen and oxygen. When heated in the absence of air, these compounds decompose to form carbon and water. If 310 g of a carbohydrate leave a residue of 124 g of carbon on heating in absence of air, what is the empirical formula of the carbohydrate ?

(Ans.: CH_2O)

N. Write each of the following in exponential notation :

a. 3,672,199 b. 0.000098

c. 0.00461 d. 198.75

O. Write each of the following numbers in ordinary decimal form :

a. 3.49×10^{-11} b. 3.75×10^{-1}

c. 5.16×10^{4} d. 43.71×10^{-4}

e. 0.011×10^{-3} f. 14.3×10^{-2}

g. 0.00477×10^{5} h. 5.00858585

P. Perform each of the following calculations. Round off your answers to two digits.

a. $\dfrac{1}{3.40 \times 10^{24}}$; b. $\dfrac{33}{9.00 \times 10^{-4}}$;

c. $\dfrac{1.4 \times 10^{9}}{(2.77 \times 10^{3})\,(3.76 \times 10^{5})}$;

d. $\dfrac{(4 \times 10^{-3})\,(9.9 \times 10^{-7})}{(789)\,(1.002 \times 10^{-10})\,(0.3 \times 10^{2})}$

Q. Perform each of the following calculations. Round off your answers to three digits.

a. (3.26 104) (1.54 106)

b. (8.39 107) (4.53 109)

c. $\dfrac{8.94 \times 10^{6}}{4.35 \times 10^{4}}$

d. $\dfrac{(9.28 \times 10^{9})\,(9.9 \times 10^{-7})}{(511)\,(2.98 \times 10^{-6})}$

R. Perform the following operations :

a. $3.971 \times 10^{7} + 1.98 \times 10^{4}$;

b. $1.05 \times 10^{-4} - 9.7 \times 10^{-5}$;

c. $4.11 \times 10^{-3} + 8.1\ 10^{-4}$;

d. $2.12 \times 10^{6} - 3.5 \times 10^{5}$.

S. A 1.000 mL sample of acetone, a common solvent used as a paint remover, was placed in a small bottle whose mass was known to be 38.0015 g. The following values were obtained when the acetone - filled bottle was weighed : 38.7798 g, 38.7795 g and 38.7801 g. How would you characterise the precision and accuracy of these measurements if the actual mass of the acetone was 0.7791 g ?

(Ans.: ±0.07736% 0.1027%)

T. Your laboratory partner was given the task of measuring the length of a box (approx 5 in) as accurately as possible, using a metre stick graduated in milimeters. He supplied you with the following measurements:

12.65 cm, 12.6 cm, 12.65 cm, 12.655 cm, 126.55 mm, 12 cm.

a. State which of the measurements you would accept, giving the reason.

(Ans.: 12.6 cm)

b. Give your reason for rejecting each of the others.

U. What weight of calcium oxide will be formed on heating 19.3 g of calcium carbonate ?
(At. wt. : Ca = 40; C = 12; O = 16)
(Ans. : 10.8 g)

V. The hourly energy requirements of an astronaut can be satisfied by the energy released when 34 grams of sucrose are "burnt" in his body. How many grams of oxygen would be needed to be carried in space capsule to meet his requirement for one day ?
(Ans. : 916.21 g)



Activity :

Collect information about various apparatus/instruments used in chemistry laboratory and make presentation of it in science exhibition.



Internet my friend

1.Error in chemical analysis
https://chem.libretexts.org>...>chapters
2. Collect information about Analytical chemistry.